

Pardhu Sri Rushi Varma Konduru

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EDUCATION

Malla Reddy University Hyderabad

Hyderabad, India

Bachelor of Technology - Computer Science & Engineering (Cybersecurity); CGPA: 7.69 August 2023 - April 2027

SKILLS SUMMARY

- **Languages:** Python, Rust, C++, JavaScript, SQL, Bash
- **Cybersecurity:** Network Security, Web Security, Vulnerability Assessment & Penetration Testing (VAPT), Threat Modeling
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, NLP, Reinforcement Learning, Trustworthy ML
- **Systems & Embedded:** Embedded Systems (ESP32), Microcontroller Programming, Firmware Basics
- **Infrastructure & DevOps:** Docker, Kubernetes, CI/CD Pipelines, Git, Linux Networking, Containerization, Virtualization
- **Cloud & Platforms:** AWS, GCP, Linux, Arduino, ESP32
- **Research:** ML Research, Embedded & Wireless Security Research

EXPERIENCE

- **Breachpoint CTF 2026** Online & Offline
Technical Challenge Author & Cloud Deployment Operations Jan 2026 - Mar 2026
 - **Security Challenge Development:** Designed challenges across Binary Exploitation, Cryptography, Reverse Engineering, Blockchain, Mobile Security, and Steganography.
 - **Low-Level Exploitation:** Engineered vulnerabilities involving memory corruption, race conditions, and adversarial binaries using C/C++ and Assembly.
 - **Reverse Engineering & Analysis:** Applied static and dynamic analysis techniques using disassembly, debugging, and binary inspection tools.
 - **Cross-Language Systems:** Developed challenges in Python, C/C++, Rust, Go, Kotlin, Solidity, and x86 Assembly to explore attack surfaces.
 - **Infrastructure & Deployment:** Deployed and managed containerized CTF infrastructure on GCP using Docker, CI/CD pipelines, and GitHub workflows for scalable and reproducible environments.
 - **Operational Reliability:** Maintained infrastructure stability and access control during live competitive environments.
- **Independent ML Research** Self-Employed
Independent ML Researcher Dec 2025 - Present
 - **Research Focus:** Investigating structural robustness, reversible behavioral learning, and parameter-efficient adaptation in neural systems.
 - **LoRA-Based Systems:** Exploring modular low-rank adaptation (LoRA) for controllable, reversible, and safety-oriented AI architectures.
 - **Theoretical & Experimental Work:** Developing frameworks for reversible adaptation, lifecycle governance, and stability analysis of learned behaviors.
 - **Publications & Prototypes:** Publishing preprints and building experimental prototypes to evaluate limitations of weight-based learning and emergent behavior control.

PROJECTS

- **RLAE Research — Runtime Low-Rank Adaptive Environments:** Exploring reversible, modular learning systems with externalized behavioral adaptation for controllable and auditable AI architectures.
- **Project Ouroboros (Ouromini) — Embedded Wireless Security Research:** Designed a modular ESP32-based system for real-time wireless monitoring, analysis, and controlled security experimentation across Wi-Fi and BLE. Developed as a dual-use research testbed for cyber-physical systems and wireless security.
- **EpsilonShift — Adversarial Fingerprint Obfuscation:** Built a research prototype injecting deterministic adversarial perturbations into browser fingerprinting APIs (Canvas, WebGL, Audio) using Python and Playwright to evaluate privacy defenses.
- **PhishVault — Containerized Phishing Analysis Platform:** Developed a scalable platform for analyzing and managing malicious URLs with automated reporting and classification. Built using Node.js, MongoDB, Puppeteer, and Docker.

PUBLICATIONS

- **On the Structural Limitations of Weight-Based Neural Adaptation and the Role of Reversible Behavioral Learning:** Pardhu Sri Rushi Varma Konduru. *arXiv preprint - arXiv:2603.02934 (cs.LG, cs.AI), 2026*. Introduces structural irreversibility in weight-based adaptation and proposes reversible behavioral learning with a Recoverability Factor for deterministic rollback. [arXiv](#) — [PDF](#)

HONORS AND AWARDS

- 2nd Place — GrowthNinja Hackathon (IIT Hyderabad, Tinkers Lab IITH) Mar 2025
- 15th Place — CY4TF Cyber Fortress CTF (Team Captain, ZenRage; 15/50 teams, 1200 pts) Mar 2025